Reg. No. | | | | | | | | | | |

[18MSCH1H03]

FIRST SEMESTER M.Sc. UNIVERSITY EXAMINATION, NOVEMBER 2019

PHYSICAL CHEMISTRY- I

Time :03 Hours **Maximum Marks:70**

Instructions
1. **Question 1 is compulsory, carries 10 marks.**
2. **Questions 2 to 7 carry 15 marks each, answer any four.**

1. Answer any FIVE of the following questions: 5×2=10

a. Define second law of thermodynamics.

b. What is the significance of chemical potential?

c. Define expectation value in quantum mechanics.

d. State the principle of NMR technique to study reaction kinetics.

e. Define molar conductance and write its relation with concentration.

f. What are the types of errors in coulometric titrations?

2. a. Derive Gibbs-Duhem and Gibbs-Helmoholtz equations.

b. Discuss the limitations of Van't Hoff equation. **10+5**

3. a. Explain the determination of fugacity of real gases by graphical method and from equation of state.

b. Derive and explain Henry's law. **10+5**

4. a. Solve Schrodinger's equation in Cartesian co-ordinate system for a particle in a 3D box.Prove that the degeneracy of the energy levels arises from symmetry.

b. Consider an electron in superfluid He (^{4}He) where it forms a solvation cavity with a radius of 18 Å. Calculate the zero point of energy and energy difference between the ground and first excited states by approximating the electron by a particle in 3D box. **10+5**

5. a. Derive kinetic equation for photochemical reaction between hydrogen and bromine using steady state approximation.

b. Explain continuous and stopped flow methods for unimolecular reaction. **10+5**

6. a. i. Define transport number. Explain the Hittorf method of determination of transport number.

ii. Calculate the ionic strength for a solution of 0.10 M NaCl and 0.10 M Na_2SO_4

b. Calculate the equilibrium constant of K^+ and SO_4^{2-} in a 0.020 M solution of K_2SO_4 with the aid of activity. **10+5**

7. a. Describe the types of instruments used for the measurement of potentials in potentiometric titrations. Give the factors that affects the potential charge.

b. Discuss dead stop and point method in biamperometric titration. **10+5**

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112